

CYCLE 1: $AC \leftarrow \overline{M[3]} + 1$

Controls: (AR=03 , BUS=110 , ALU=0111 , AC_Load=1 , AC_INC=1)

***this will complement $M[3]$ and add 1 to get $-M[3]$ and stores it in AC*

CYCLE 2: $AC \leftarrow AC * M[0]$

Controls: (AR=00 , BUS=110, ALU=1001, AC_Load=1)

***this is $AC = (-M[3]) * M[0]$*

CYCLE 3: $M[4] \leftarrow AC$

Controls: (AR=04 , BUS=011 , ALU=0000, Store=1)

***this will store the value of AC in $M[4]$*